



UNIVERSITEIT VAN PRETORIA
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YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en
Inligtingtegnologie / Lefapha la Boetšenere,
Tikologo ya Kago le Theknolotši ya Tshedimošo

Unit 13: Systems Thinking & Business Structure Integratedness

BSS 310 Engineering Management 2026

Lecturer: Prof. Michael Ayomoh



Quick Recap/Questions arising from Previous Lecture

The lecture addressed:

- Definitions and introductory rudiments of cash flow diagrams
- Discussions on time value of money in general (with more focus on discounting to the present in a bid to obtain the present value (PV) of money)
- A practical relationship between the time value of money, the MARR and a Project's viability
- A linkage of the theoretical discussions of the time value of money with the semester project

Learning Outcomes for the Current Lecture

The expected learning outcomes include:

- **Understanding** the fundamentals of systems thinking.
- **Analysing** and **Evaluating** the diversity of causal loop diagrams in systems thinking.
- **Creating** illustrative sample models via demonstration of the systems thinking process with the use of the Vensim software
- **Application** of the entire concept to the semester project.

Presentation Outline

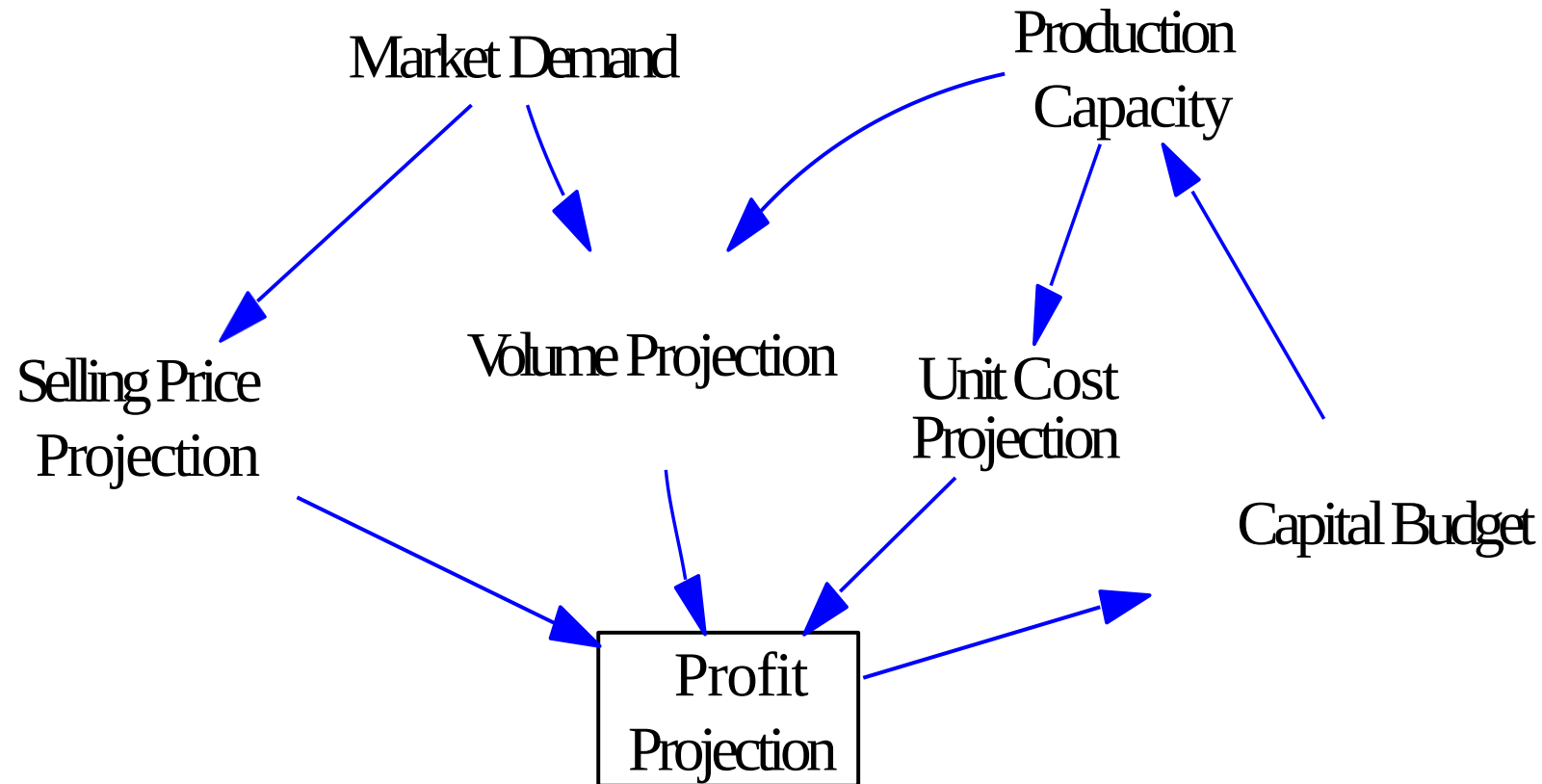
- Definition of Systems thinking and the relevance of Systems thinking in:
 - Project Management or
 - Business Systems Design and Development
- System Behaviour and Structure
- System Structure & Causal Loop Diagram
- Application of acquired knowledge to the Semester Project
 - Establish a systems thinking connectivity for the two or three portfolios in your business
- Vensim Demo

Definition of Systems Thinking: **What is Systems Thinking**

It is an approach towards understanding a problem from a holistic perspective by considering all interacting entities and the relationship that exist amongst them.

- The method of systems thinking provides us with tools for better understanding difficult problems. However, this approach requires a shift in the way we think about the performance of a system.
- In particular, it requires that we move away from looking at isolated events and their causes (usually assumed to be some other events), and start to look at the system made up of interacting system elements.

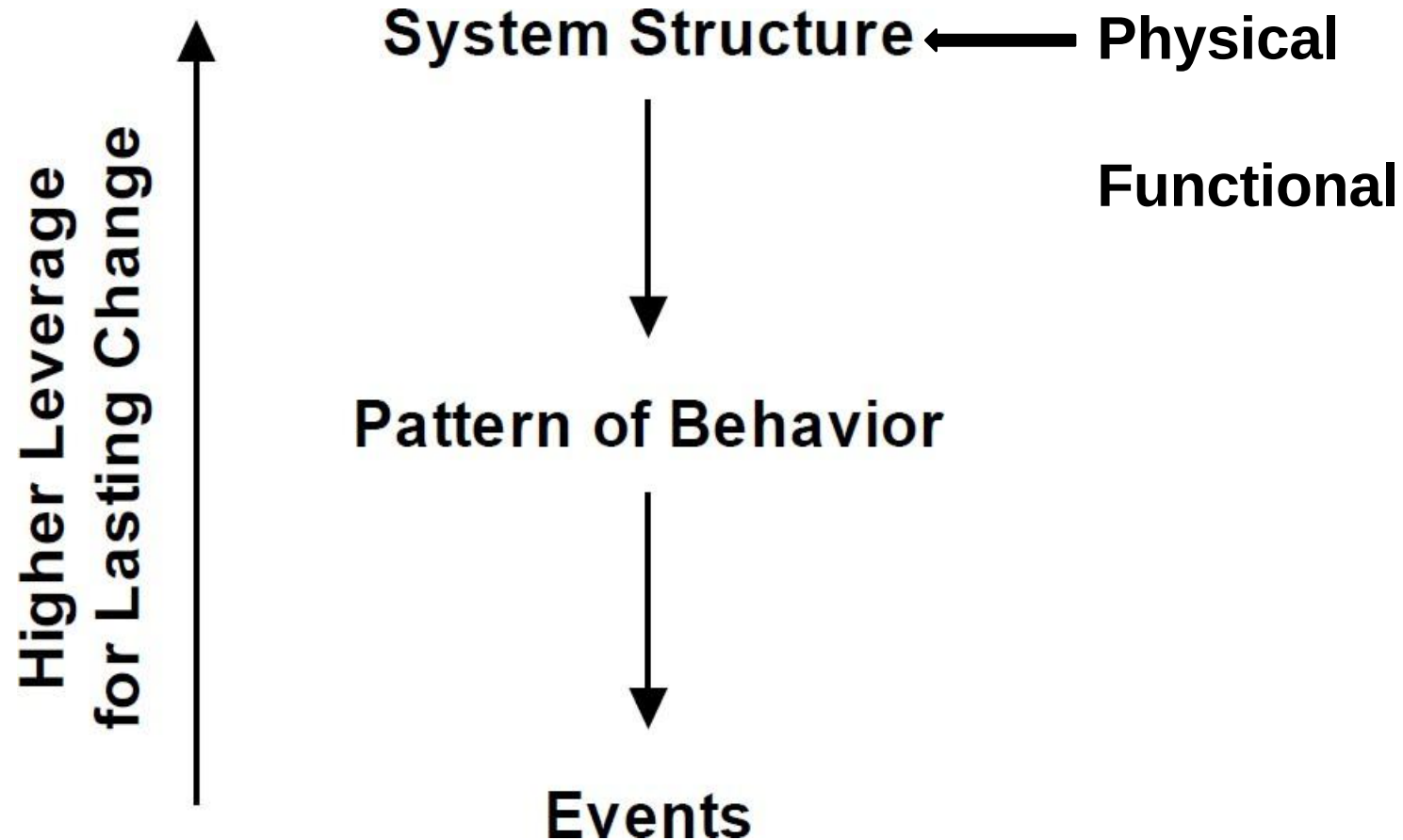
Example of an interrelationship and a Cyclic System Structure in a Project or Business Systems



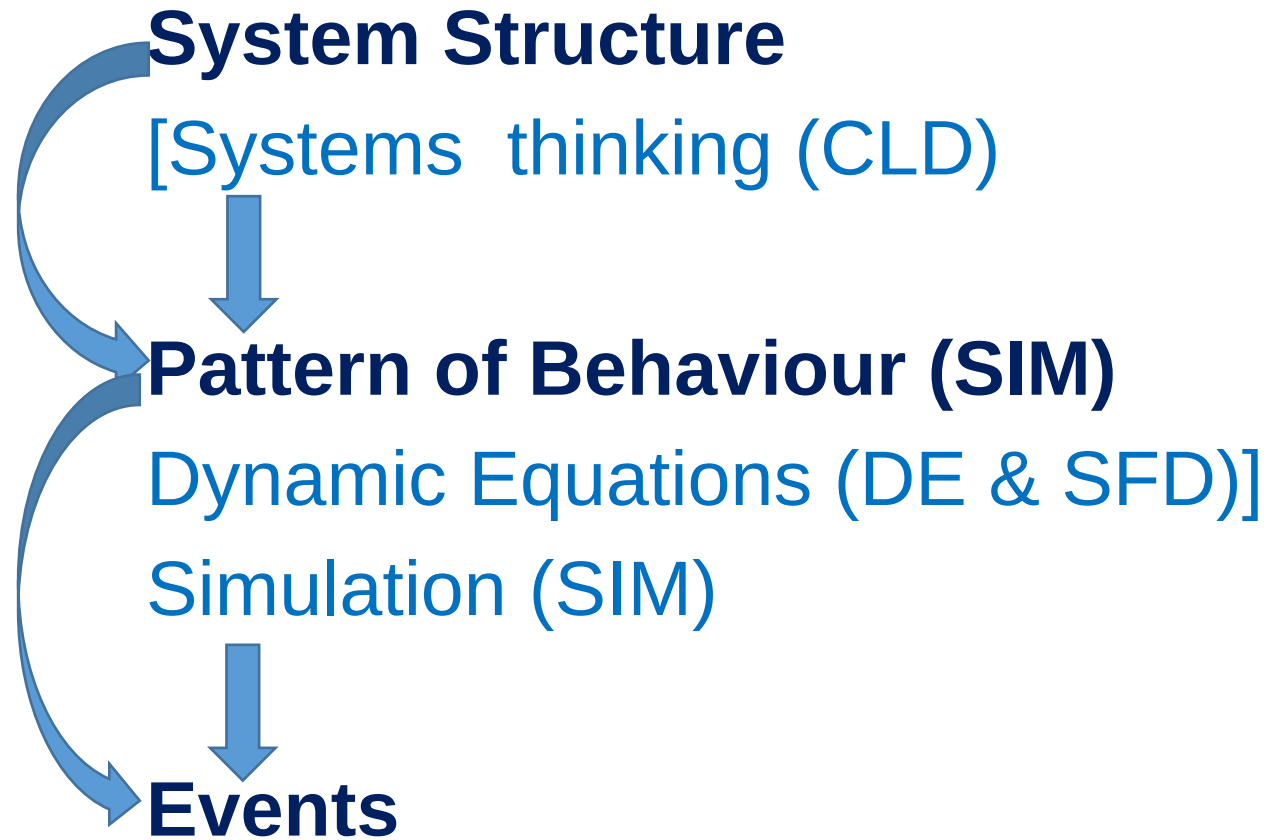
System Behaviour & Structure

- All systems exhibit some form of behaviour.
- Some behaviours are fixed, while others are periodic or transient with time, through the influence of other driving factors.
- In a bid to address problems posed to us as a result of the exhibited systemic behaviours, **humans are quick to drawing conclusions on an isolated troubled event as the problem and another event as the source of the problem.**

Holistic Picture of Systemic Problems Emergence



Hierarchical Levels in (Systems Thinking & Dynamics) Modelling



Legend:

CLD-Causal Loop Diagram

SFD-Stock & Flow Diagram

SIM- Simulations

DE-Dynamic Equations

System Structure & Causal Loop Diagram

Causal Loop Diagram

To better understand the system structures which cause the patterns of behavior earlier discussed, a notation for representing system structures is discussed below.

The usefulness of a graphical notation for representing system structure is illustrated using the Causal Loop Diagram (CLD). The CLD shows:

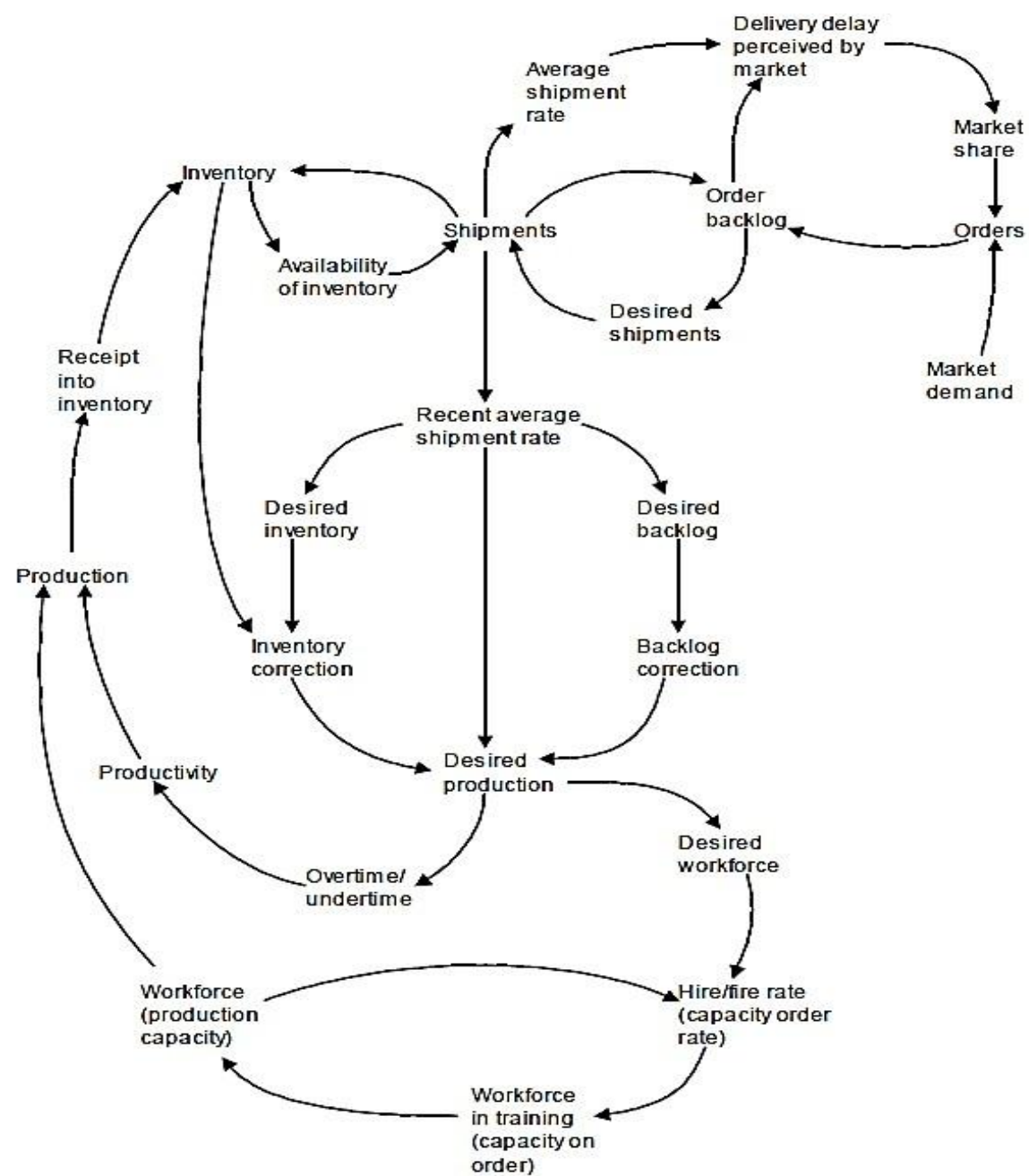
- i. the relationships among the elements of a system.

Causal Loop Diagram

ii. The CLD presents relationships that are difficult to verbally describe because normal language presents interrelations in linear cause-and-effect chains, while the diagram shows that in the actual system there are possible circular chains or closed cause-and-effect linkages.

Causal Loop Diagram

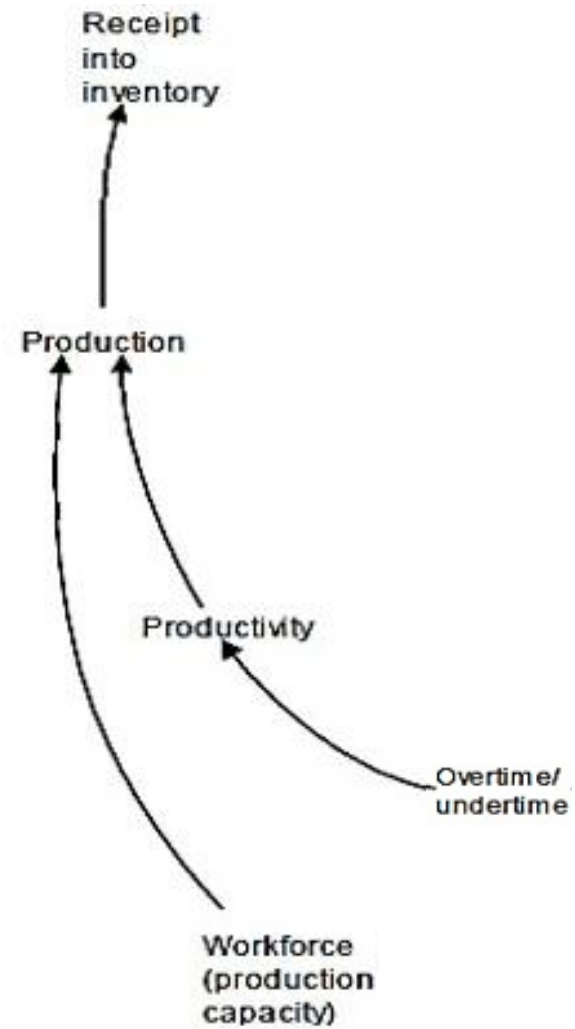
- When an element of a system indirectly influences itself it is called **a feedback loop** or a causal loop. Feedback was defined as the transmission and return of information (Richardson and Pugh 1981).] More formally, a feedback loop is a closed sequence of causes and effects
- The feedback structure as presented below, consist of elements and arrows (which are called causal links) linking all elements together to make a system.



Feedback structure of a basic production sector

Causal Loop Diagram

A linear chain of causes and effects which does not close back on itself is called **an open loop**. An analysis of causes and effects which does not take into account feedback loops is sometimes called open loop thinking



Feedforward structure of a basic production sector

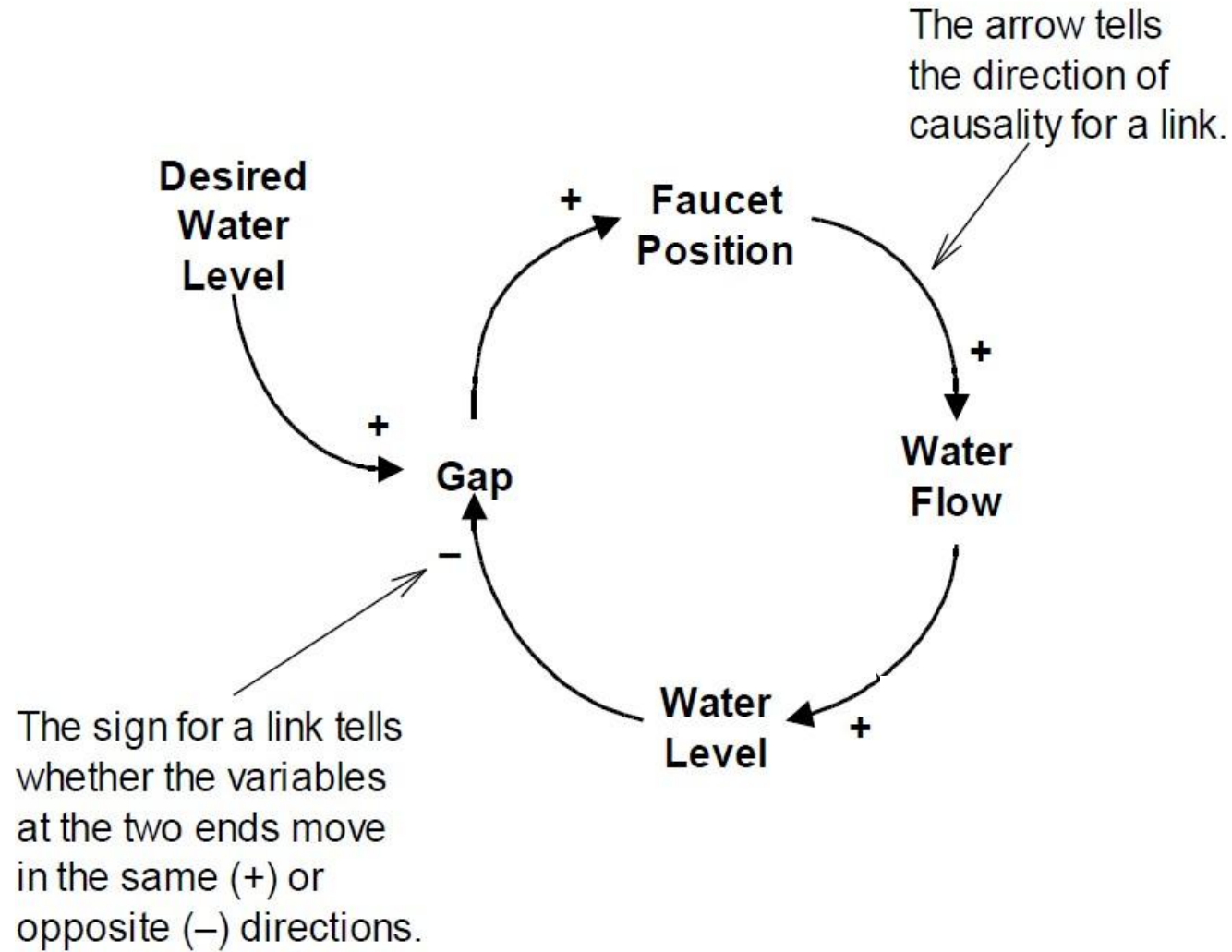
Causal Loop Diagram (Link sign)

However, additional information aids with a more complete analysis. This includes the use of signs (either + or -) on each link. These signs have the following meanings:

- a) A causal link from one element A to another element B is positive i.e. (+) if either (i) A adds to B or (ii) a change in A produces a change in B in the same direction.

- b) A causal link from one element A to another element B is negative i.e. (-) if either (i) A subtracts from B or (ii) a change in A produces a change in B in the opposite direction. **See an example below**

CAUSAL LOOP DIAGRAM [Filling a glass of water]



Filling a glass of water (with both the Causal link signs & Loop sign)

Causal Loop Diagram (Alternative Link signs “s” or “o”)

An alternative notation is used in some presentations of causal loop diagrams. With this alternate notation, a lower case s is used instead of a (+) on a link, and a lower case o is used instead of a (-). The s stands for “same” and the o stands for “opposite” indicating that the variables at the two ends of the link move in either the same direction (s) or opposite directions (o).

Causal Loop Diagram (Alternative Link signs “R” or “B”)

In some loops, a capital R is used instead of (+), and a capital B is used instead of (-). The R stands for “reinforcing,” and the B stands for balancing.”

Causal Loop Diagram (Loop Sign)

In addition to the signs on each link, a complete loop also is given a sign. The sign for a particular loop is determined by counting the number of minus (-) signs on all the links that make up the loop i.e.

- i) A feedback loop is called positive, indicated by a (+) sign in parentheses, if it contains an even number of negative causal links.
- ii) A feedback loop is called negative, indicated by a (-) sign in parentheses, if it contains an odd number of negative causal links.

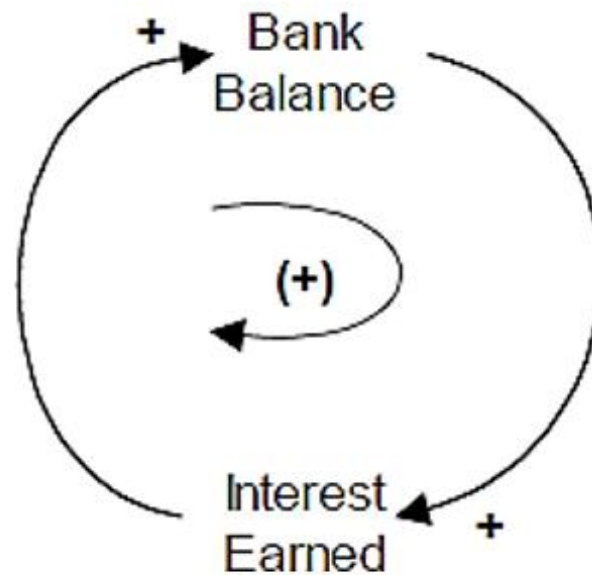
Causal Loop Diagram (Loop Sign contd...)

Thus, the sign of a loop is the algebraic product of the signs of its links. Often **a small looping arrow is drawn around the feedback loop sign** to more clearly indicate that the sign refers to the loop.

See examples below:

Positive (Reinforcing) Feedback Loop

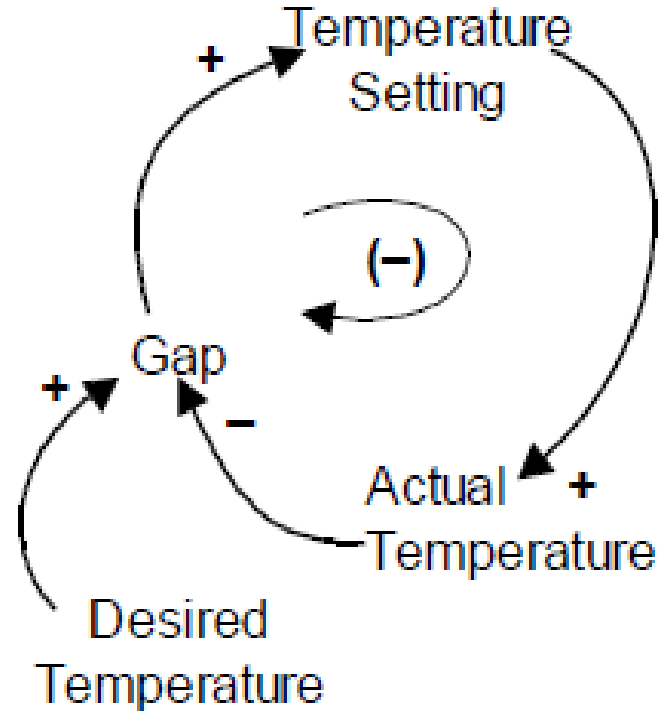
System Structure



Positive feedback loop: Growth of bank balance

Negative (Balancing) Feedback Loop Contd.....

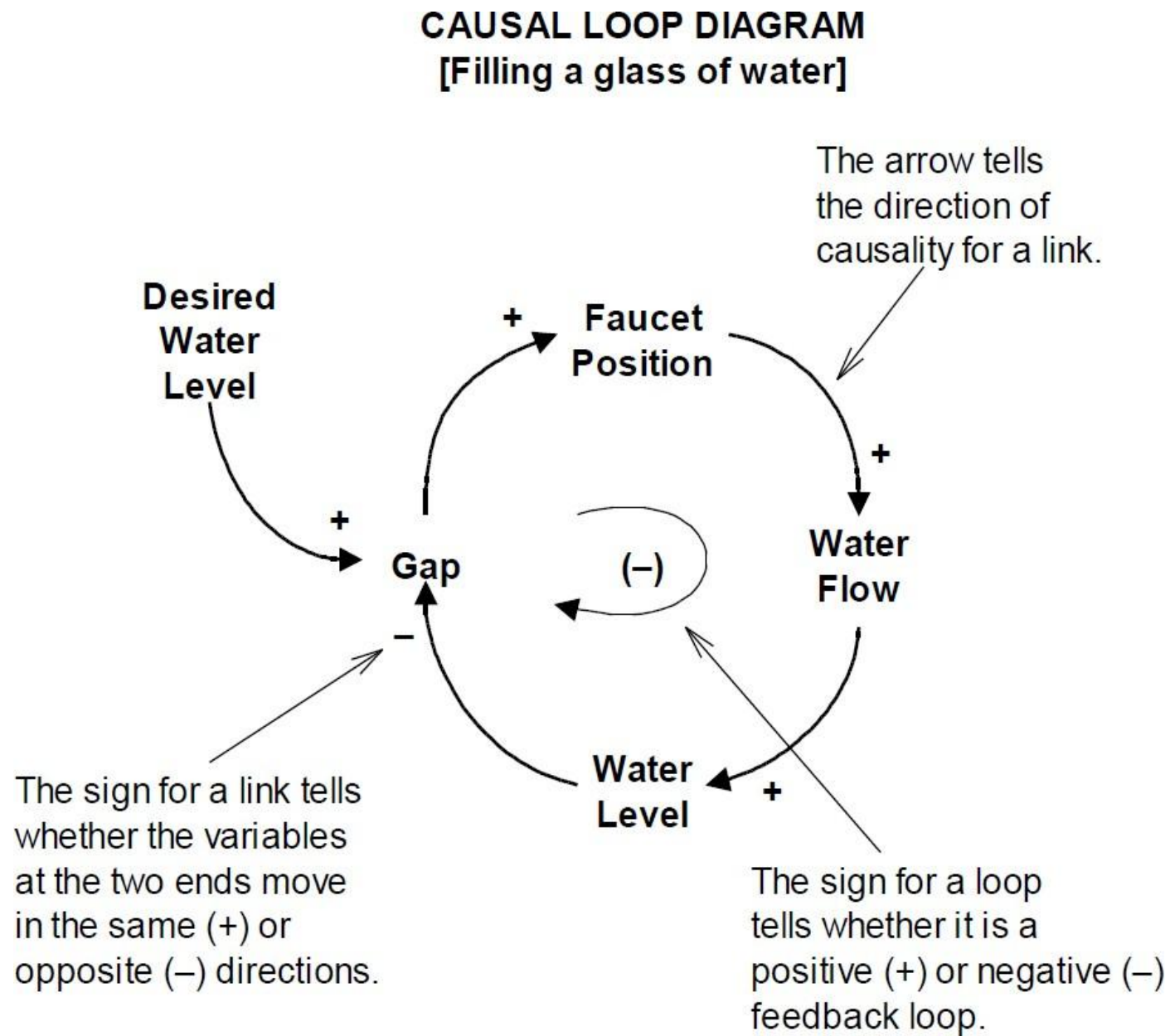
System Structure



A negative feedback loop diagram for the regulation of an electric blanket temperature.

Negative(Balancing)Feedback

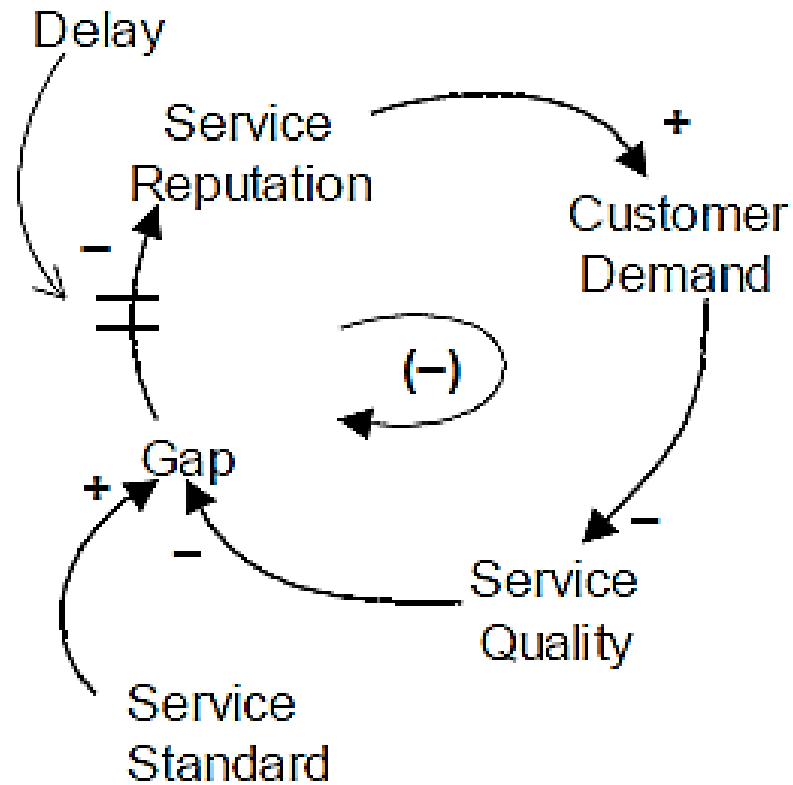
LoopContd.....



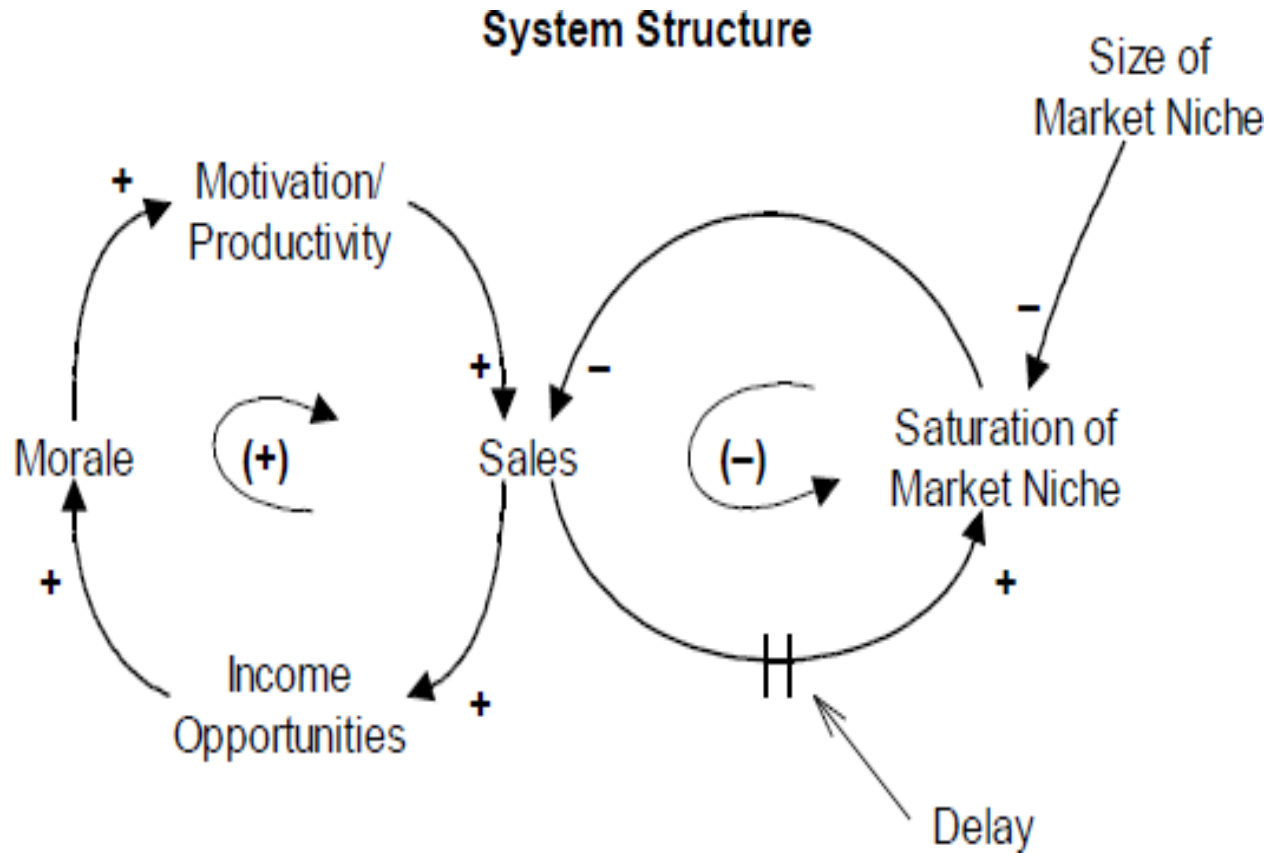
Filling a glass of water (with Causal link signs)

Negative Feedback Loop with Delay

System Structure



Combination of Positive and Negative Loops



Creating Causal Loop Diagrams

To start drawing a causal loop diagram, decide which events are of interest in developing a better understanding of system structure.

Creating Causal Loop Diagrams

The following hints for drawing causal loop diagrams are based on guidelines by Richardson and Pugh (1981) and Kim (1992):

- i) Think of the elements in a causal loop diagram as variables which can go up or down, but don't worry if you cannot readily think of existing measuring scales for these variables.
- **Use nouns or noun phrases to represent the elements, rather than verbs** i.e., the actions in a causal loop diagram are represented by the links (arrows), and not by the elements. For example, use “cost” and not “increasing cost” as an element.

Creating Causal Loop Diagrams

- Generally, use an element name for which the positive sense is preferable e.g. use "Growth" rather than "Contraction."
- **Causal links should imply a direction of causation**, and not simply a time sequence. That is, a positive link from element A to element B does not mean first A occurs and then B occurs." Rather it means, when A increases then B increases."

Creating Causal Loop Diagrams

- ii) As you construct links in your diagram, think about possible unexpected side effects which might occur in addition to the influences you are drawing. As you identify these, decide whether links should be added to represent these side effects.

- iii) **For negative feedback loops, there is a goal.** It is usually clearer if this goal is explicitly shown along with the “gap” that is driving the loop towards the goal.

Creating Causal Loop Diagrams

iv). A **difference between actual and perceived states** of a process can often be important in explaining patterns of behavior. Thus, it may be important to include causal loop elements for both the actual value of a variable and the perceived value. In many cases, there is a lag (delay) before the actual state is perceived. For example, when there is a change in actual product quality, it usually takes a while before customers perceive this change.

Creating Causal Loop Diagrams

v) There are often **differences between short term and long term consequences of actions**, and these may need to be distinguished with different loops. **For example, the short term result of taking a mood altering drug may be to feel better, but the long run result may be addiction and deterioration in health.**

vi) **If a link between two elements need a lot of explanation, you probably need to add intermediate elements between the two existing elements that will more clearly specify what is happening.**

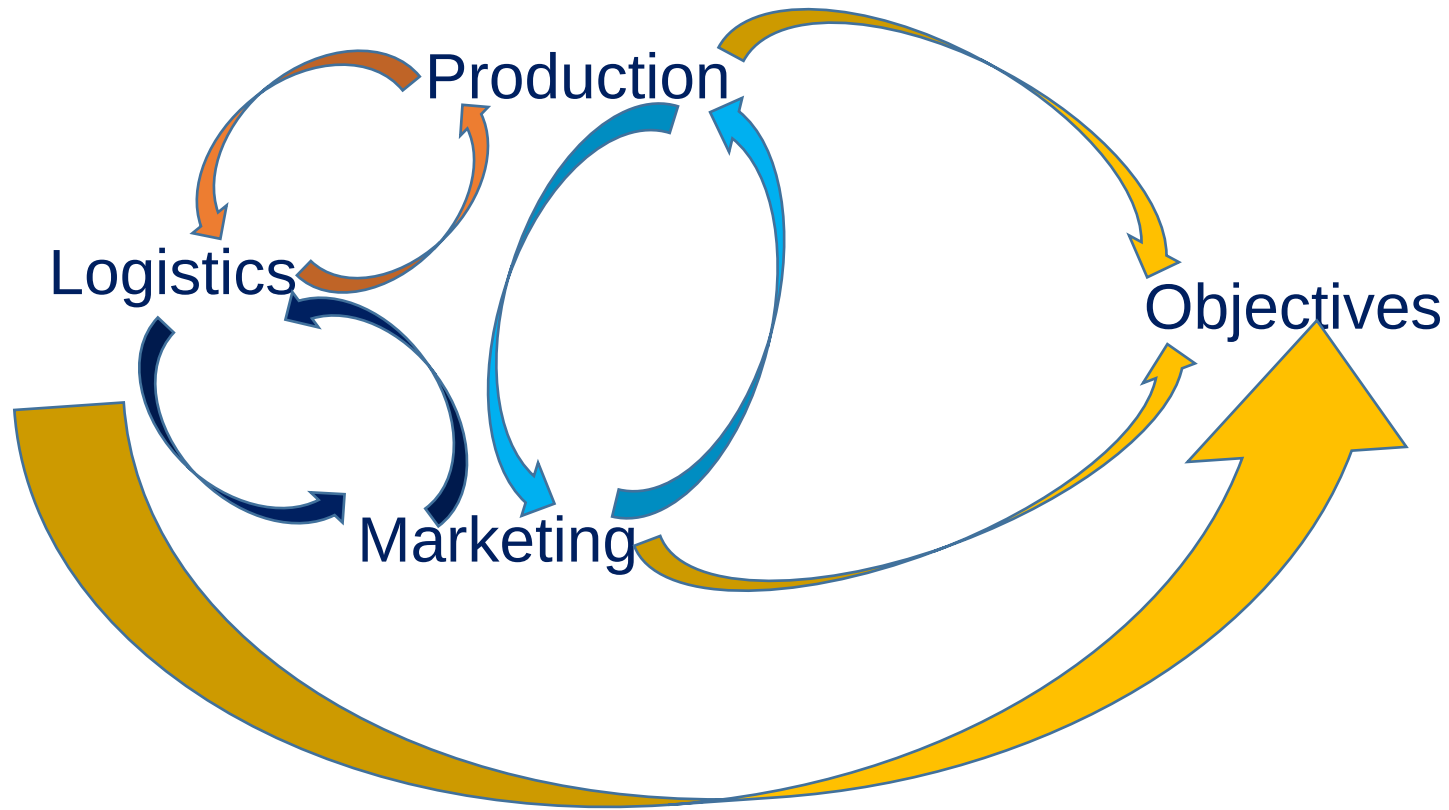
Creating Causal Loop Diagrams

vii). **Keep the diagram as simple as possible**, subject to the earlier points. The purpose of the diagram is not to describe every detail of the system's process, but to show those aspects of the feedback structure which can lead to the observed pattern of behaviour.

Application to the Semester Project

- Section 5 (PART A) (BSDM) • Systems thinking & business structure integratedness

Semester Project Cyclic Structure & Systems Thinking



Vensim Demo

Vensim In-class Sample Application

Learning Outcomes Evaluated

- Do you understand the systems thinking process?
- Can you analyse and evaluate a diverse range of causal loop diagrams in systems thinking?
- Can you construct a systems thinking model depicting integratedness of the member elements of a project or business system?
- Can you apply the systems thinking modelling approach to the different departments of your semester project?

The End

For further reading: Section covered in e-book include:

- John D. Sterman, (2000) Business Dynamics: Systems Thinking and Modeling for a Complex World, Publisher: Irwin McGraw-Hill (Part 2, Chapter 5)
- System Dynamics Methods: A Quick Introduction by Craig W. Kirkwood (Chapter One)

Next Topic

Unit 14: Business intelligence, risk management and scenarios



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